Lecture 17: Virtualization

Some slides borrowed from:
Jaswinder Pal Singh (Princeton), Harsha V. Madhyastha (Michigan), UC Berkeley CS162, and Branden Ghena (Northwestern),

- **Virtualization**

- Approaches
  - Emulation
  - Hypervisors
  - Containers

Virtualization

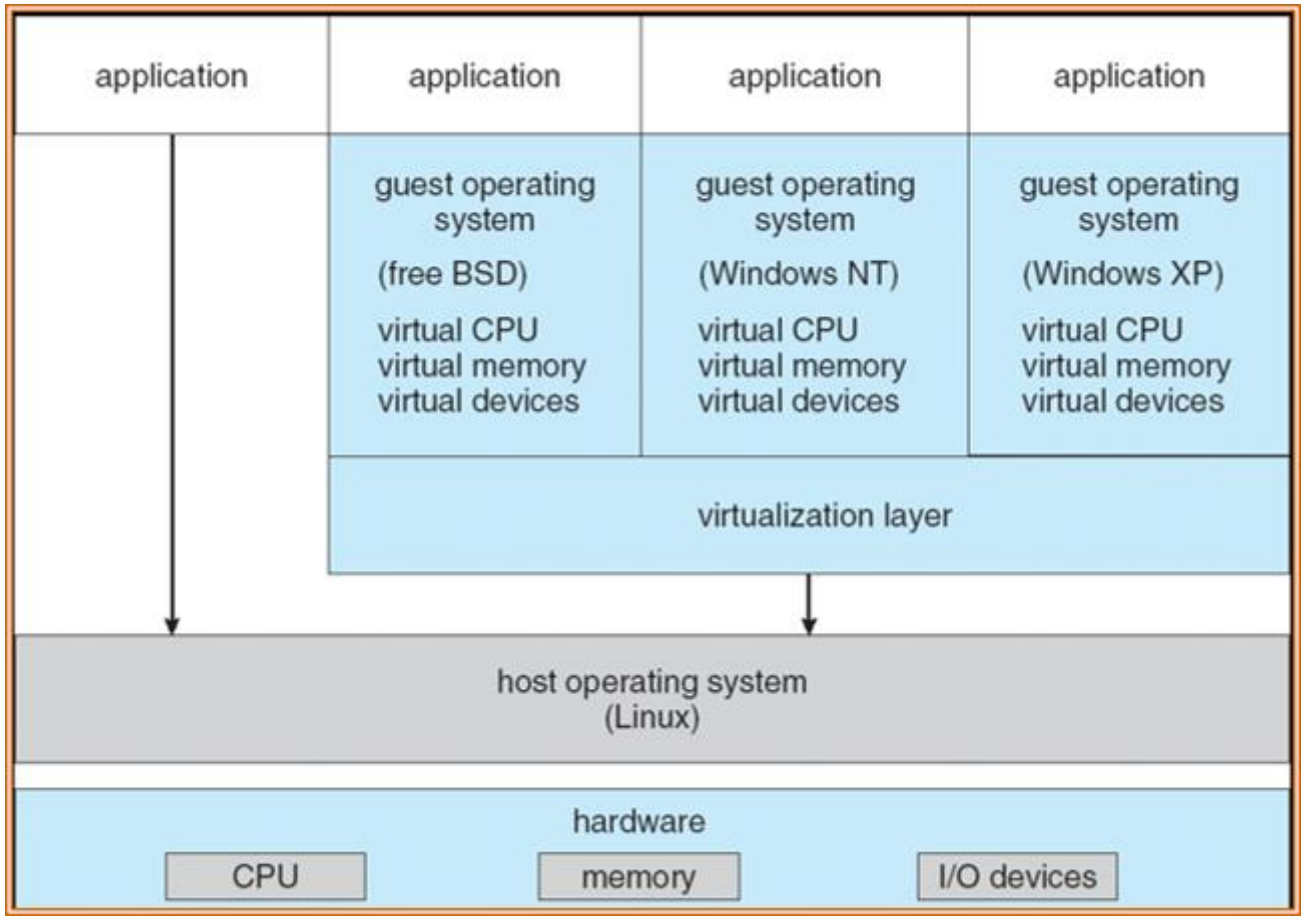
- Virtual versions of real resources are used for protection and limitation
  - Memory – virtual memory
  - CPU – processes and scheduler
  - Disk – file systems

- OS provides these abstractions to simplify applications
  - And provide security

Virtual Machines

- What about virtualizing the whole computer?
  - Provide interfaces that look like a normal computer
  - But actually interact with software that manages access

- Run an entire OS within an OS



Original motivation: support more applications

- 1960s IBM mainframes had many different OSes
  - Some applications only written for certain OSes though

- Virtualization allowed multiple OSes to run on a single mainframe
  - Which let one powerful computer serve varied needs of many people

- Still applies today to some degree
  - Windows + Ubuntu VM
    - Want PowerPoint and also terminal environment (vim/make/gcc)

  - MacOS + Windows VM
    - Various Windows-only programs
    - Often dual boot rather than VM

Modern motivation: package and isolate applications

- High-performance applications aren't really standalone
  - Assumptions about OS
  - Assumptions about libraries and services
  - Multiple processes working together

- A virtual machine is a method to encapsulate "entire stack"
  - Even down to expectations of hardware

- Cloud computing platforms run many applications together
  - Need isolation from each other in a strongly controllable way
    - Exactly 2 GB of RAM go to this
    - Exactly two processor cores go to that

Virtualization approaches

1. Simulate everything in the computer completely
   - Emulation

2. Simulate parts of the computer, but not all of it (really use CPU)
   - Hypervisor

3. Simulate the operating system (software environment)
   - Containers

- Virtualization

- **Approaches**
  - **Emulation**

  - Hypervisors

  - Containers

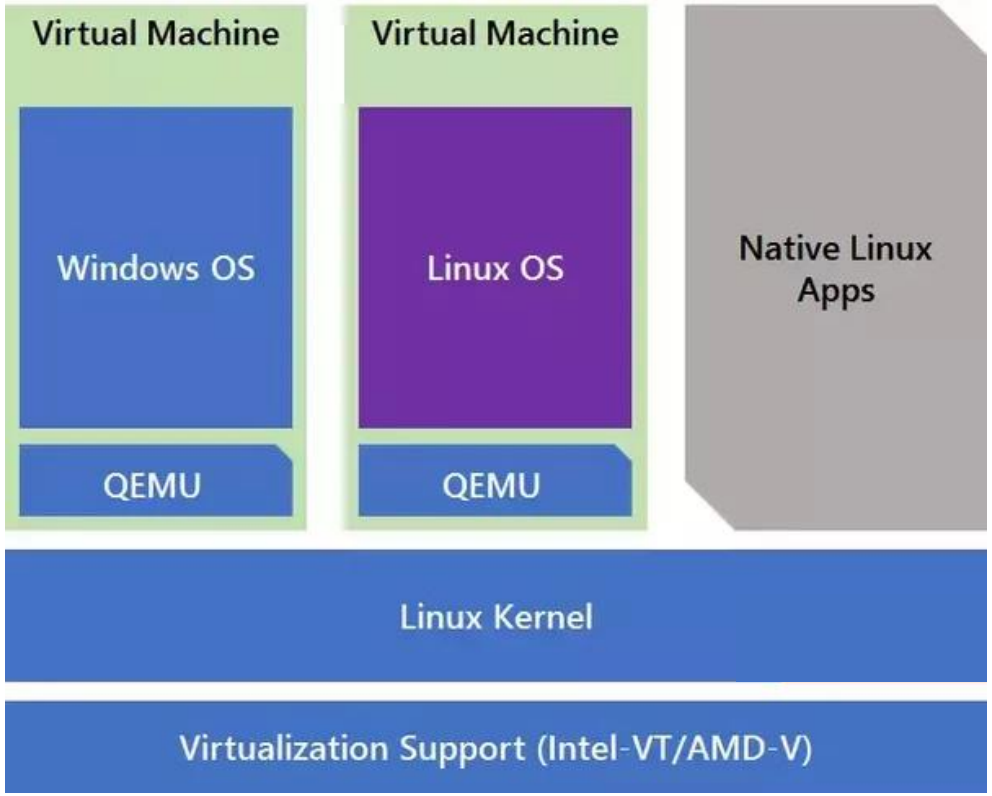
Software emulation

- User software emulates the behavior of every single instruction
  - Data structures for Processor, Memory, I/O, etc.
  - Code for Instruction Cycle:
    - Fetch next instruction
    - Decode
    - Perform operation
    - Update state (e.g., put results into some register but it’s actually the memory)

- Example: Gameboy emulator
  - Simulates every behavior as-if it were actually Gameboy hardware

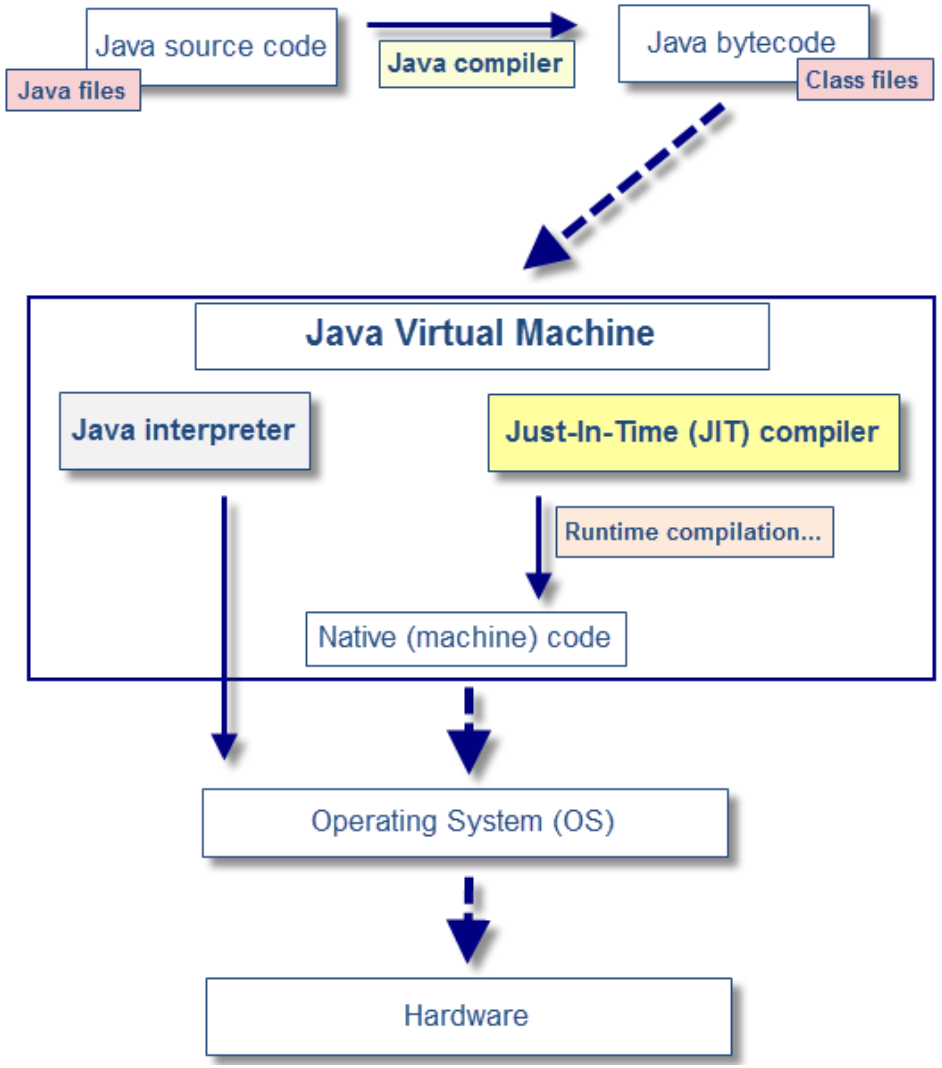
Emulation tradeoffs

- Upsides
  - Any hardware you want
  - Entirely in userspace
- Downside
  - Slower than real hardware by definition



Simple emulators: interpreted languages

- Create a simple environment for code to execute within
- Interpret code instructions (bytecode or lines of code) and perform actions
  - Example: fakes a machine that executes Java bytecode
- Still ties in to many parts of the real machine
  - Filesystem
  - Devices



Not-quite-emulation: binary translation

- MacOS on ARM (M1)
  - Uses ARM processor with ARM instruction set
  - Old programs were compiled for x86-64 instruction set
- Solution: translate assembly instructions
  - Ahead of time (AoT)
  - Or just-in-time (JIT)
  - Works fine for applications that are I/O bound
- Simulates a different CPU, but leaves the remainder of the computer the same

- Virtualization

- **Approaches**
  - Emulation
  - **Hypervisors**
  - Containers

How do we speed up virtual machines?

"Efficiency … demands that a statistically dominant subset of the virtual processor's instructions be executed directly by the real processor, with no software intervention…"

—Popek and Goldberg, 1974

- Need to use some parts of the computer for real while simulating other parts

Virtual Machine Monitor (VMM)

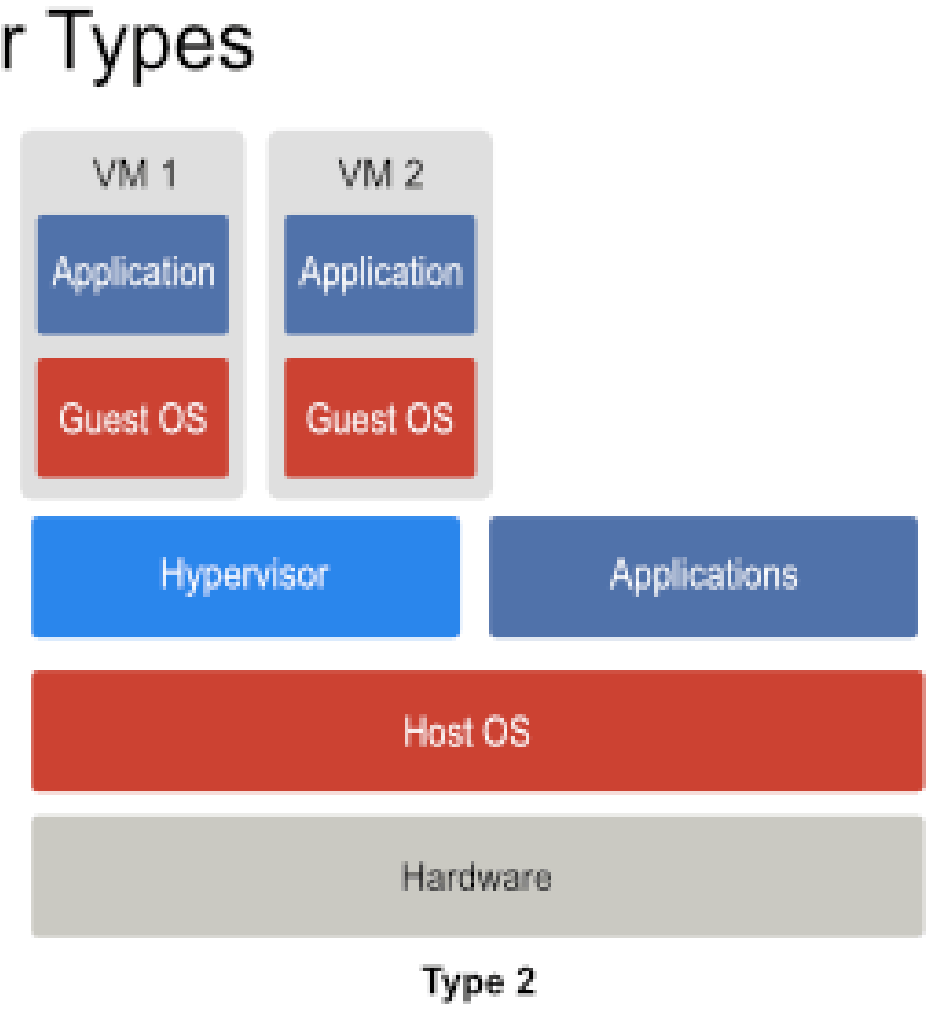
- Also known as hypervisors
  - OS kernel is the system “supervisor” and manages the computer
  - Hypervisor manages supervisors

- Creates the illusion that the OS has full control over the hardware
  - And even gives real (limited) access to hardware whenever possible
  - But may actually be sharing full computer resources among several OSes

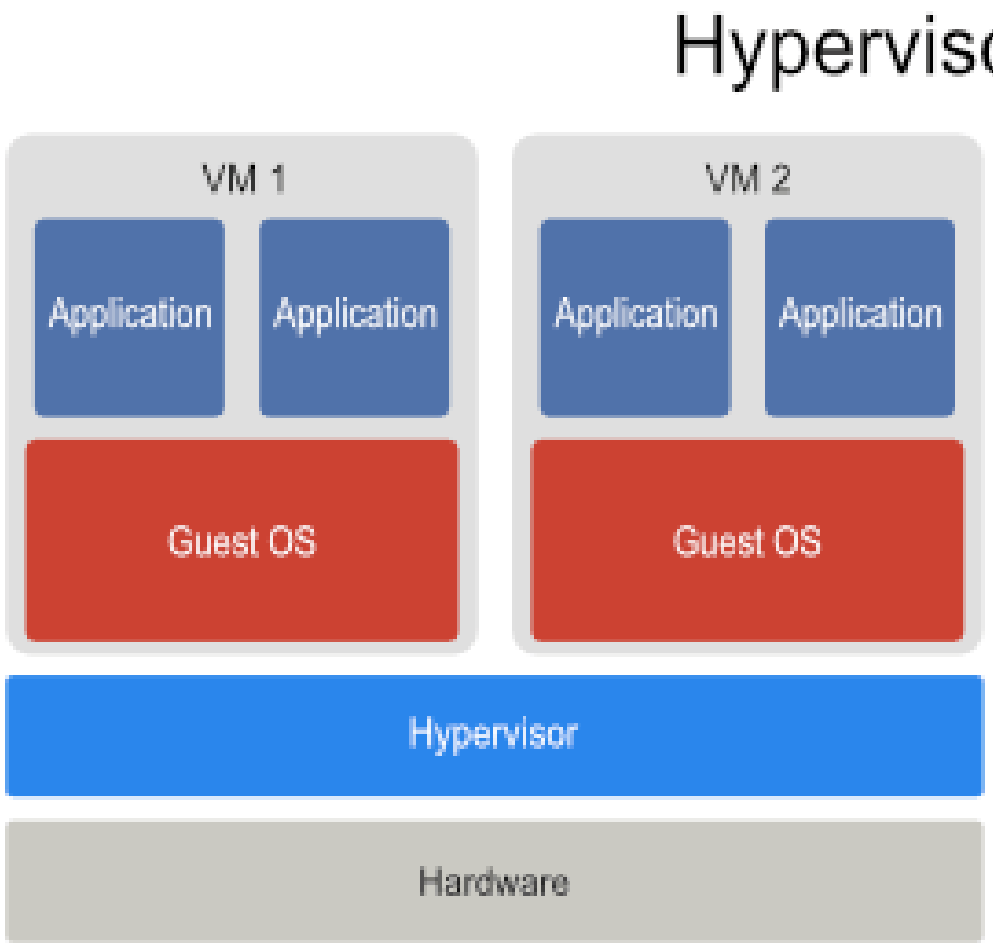
- Probably what you had in mind as virtual machines
  - VirtualBox, VMWare, Parallels

Hypervisor layering

- Normal operating system runs on hardware
  - Known as "Host OS"
- Hypervisor runs on top of host and coordinates with it to enable interactions with hardware
  - Some coordination may be within the kernel itself



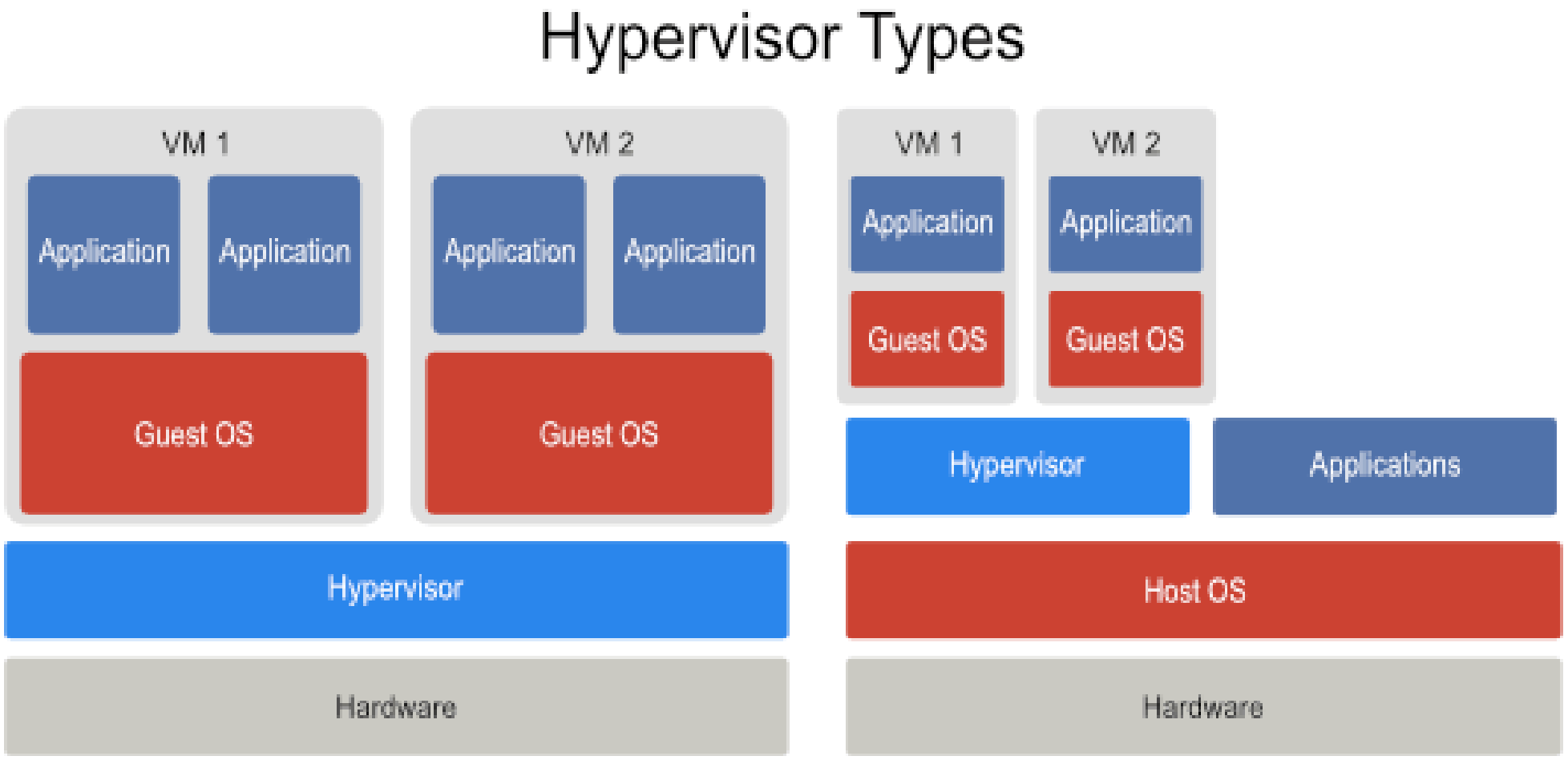
Hypervisor layering



Type 1
"Bare Metal" Hypervisor

- Hypervisor manages hardware directly
- Guest OSes run on top of it

Hypervisor layering



Type 1
"Bare Metal" Hypervisor

Type 2
Hosted Hypervisor

Abstraction choices for hypervisor

- Fully virtualizing hypervisor
  - Virtual machine looks exactly like a physical machine
    - Though not necessarily the same machine it's running on
  - Guest OS does not need to be modified in any way
  - Guest may not even be aware it's running virtually

- Para-virtualizing hypervisor
  - Guest OS has extensions to cooperate with hypervisor
  - Sacrifice transparency for better performance
  - Might include an API to interact with hypervisor
  - Guest OS likely can skip some stuff the hypervisor handles instead

Paravirtualization

- In pure virtualization the VMM intercepts and translates privileged instructions
- “Paravirtualization” means that the guest OS is modified to run on a VMM
- In paravirtualized VM, the guest OS can issue “hypercalls” to the hypervisor
  - Hypercalls are the syscall equivalent. i.e., application:syscall :: kernel:hypercall
- Paravirtualization supported by Windows and Linux

Arbitrary combinations of these are possible

	Bare Metal Hypervisor	Hosted Hypervisor
Fully Virtualized	VMWare ESXi, Microsoft Hyper-V	VMware Workstation, VMware Fusion, Parallels, VirtualBox
Para Virtualized	Xen	User Mode Linux

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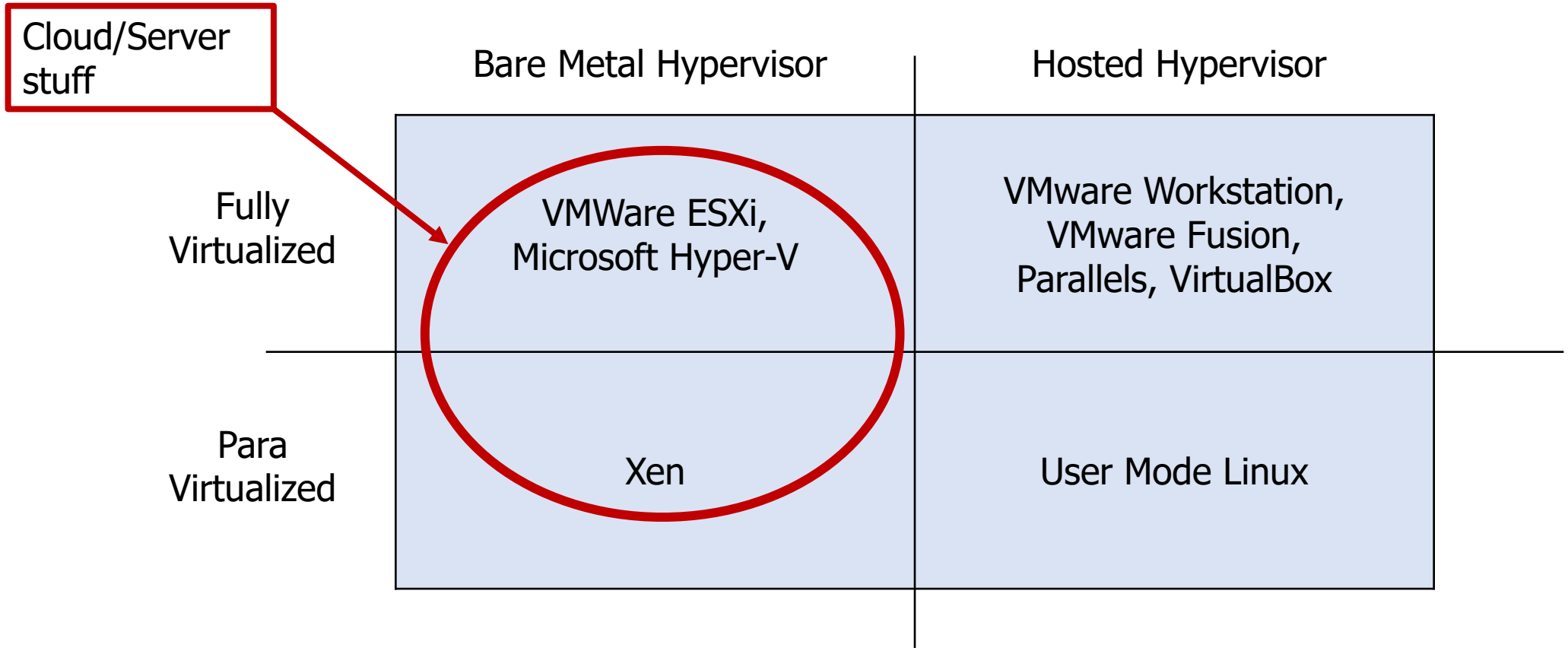
"Normal" VMs

Arbitrary combinations of these are possible



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Para Virtualized	Xen	User Mode Linux

Run Linux as a user-space app on Linux. The guest kernel uses the host kernel underneath. You have to break two kernels, not one.

Hypervisor example: system call

Process	Guest OS	Hypervisor
1. System call: trap to OS		
		2. Receive trap. Call guest OS trap handler
	3. OS trap handler: Decode trap and execute syscall. When done issue return-from-trap	
		4. OS tried to return from trap. Do real return-from-trap
5. Resume execution		

- Virtualization

- **Approaches**
  - Emulation
  - Hypervisors
  - **Containers**

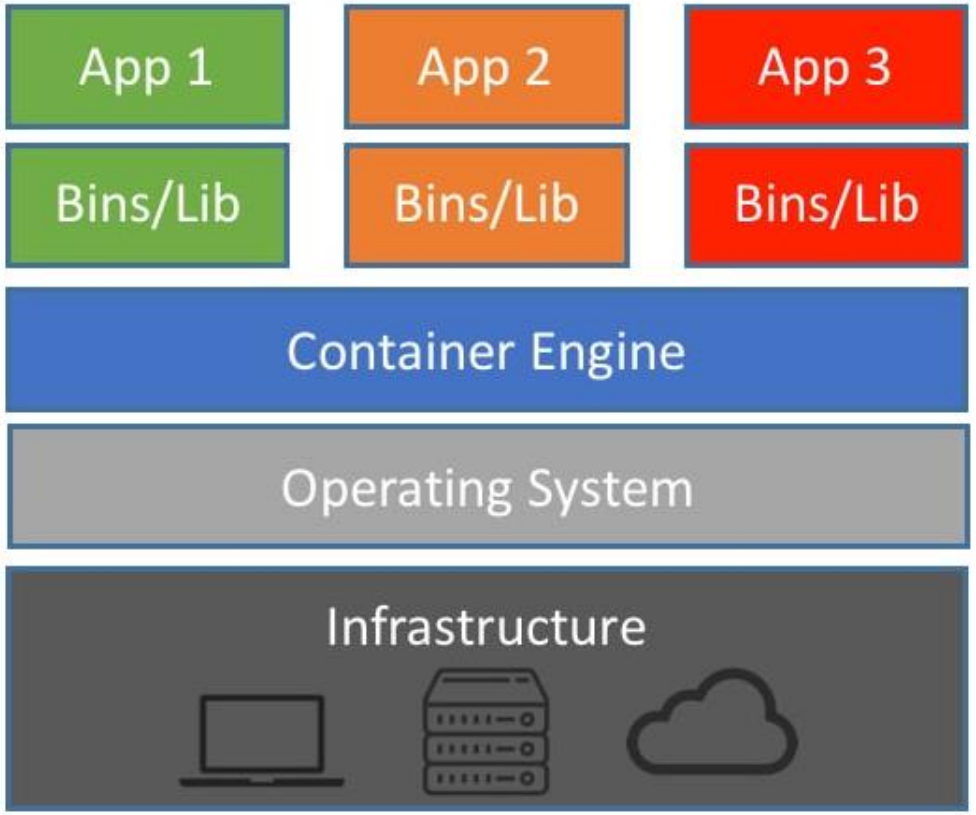
Cloud platform requirements

- May want to provide multiple OSes, but can do so with multiple physical machines

- Really want encapsulation and isolation
  - Encapsulation
    - Include particular shared libraries that application needs
    - Without interfering with other applications on system
  - Isolation
    - Guarantee certain processing and memory allocations to each application
    - Limit visibility into the filesystem (without overhead of partition per app)

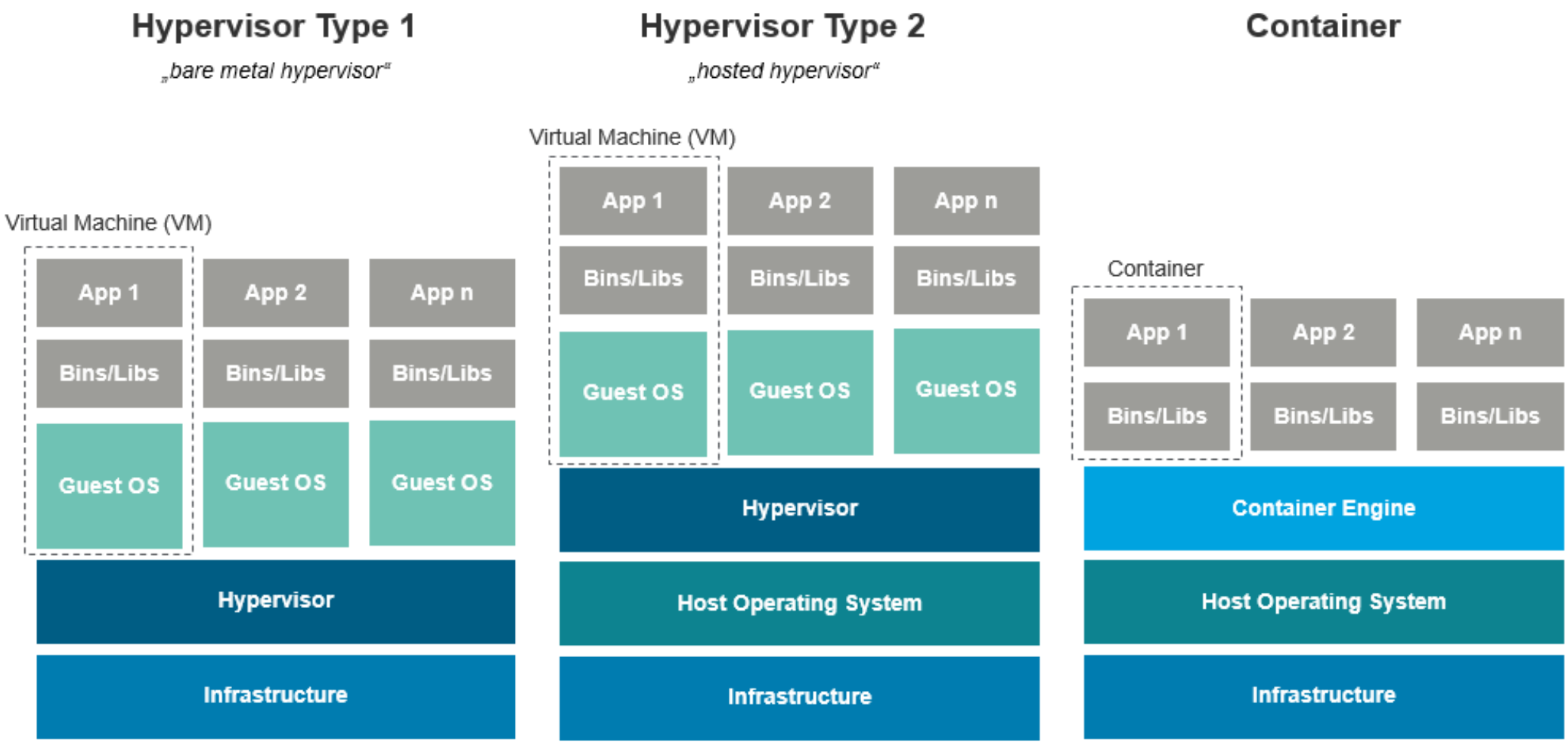
Containers

- Provide each application with illusion of its own dedicated OS
  - Isolated resources: processor and memory
  - Isolated namespace: PIDs, network, filesystem
  - Includes only the binaries and libraries it needs



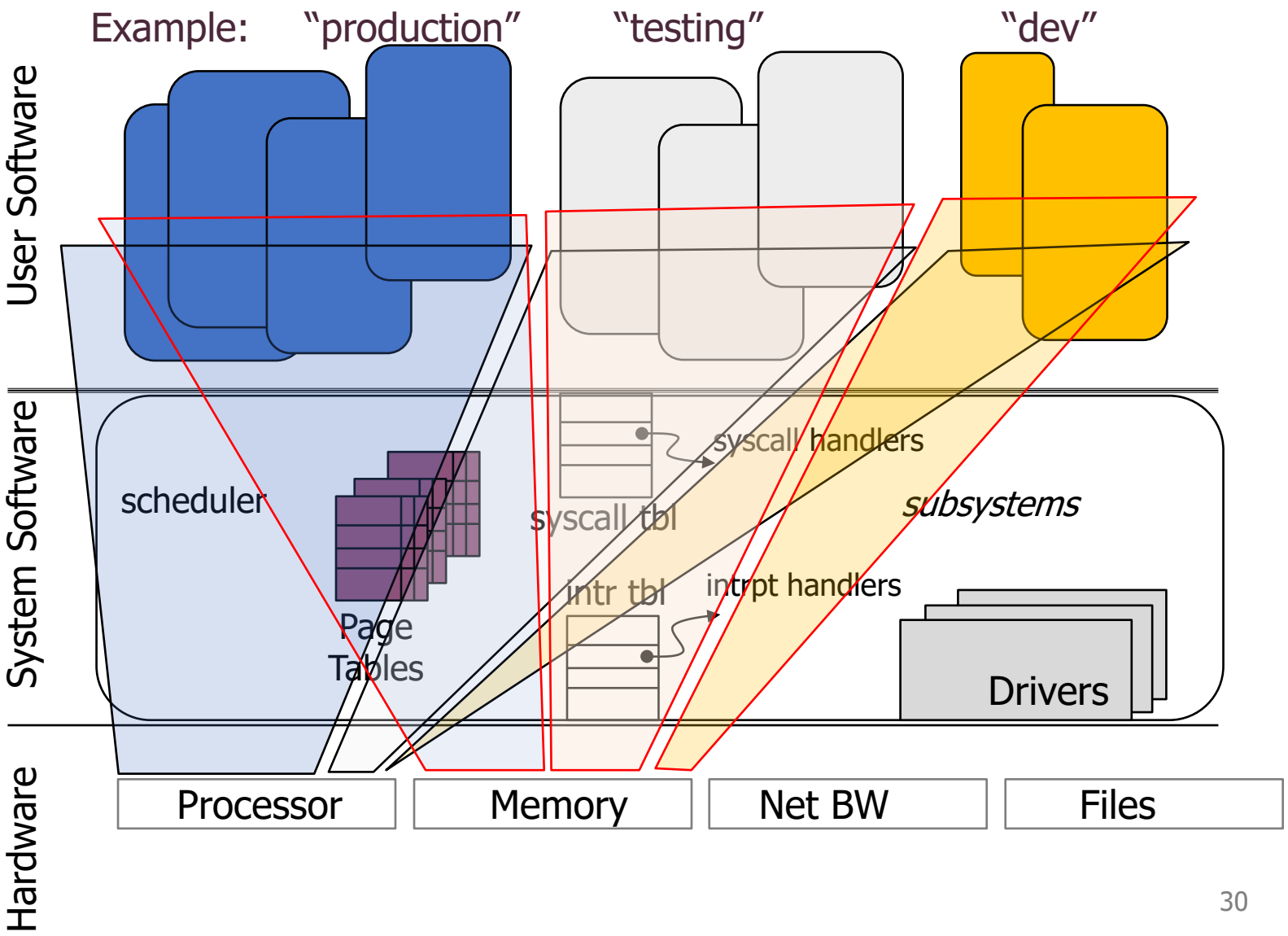
Containers

Visual comparison of Hypervisors and Containers



Linux cgroups (control groups)

- Collection of processes treated as a group for resource allocation
- Provider greater performance isolation between cgroups than between processes
  - Firm limits per group



Cgroups can be used to build containers

- Devices can be connected or denied to a cgroup
  - Cgroup processes will not be able to detect device at all

- Accounting can be done on cgroup usage
  - Memory, CPU, disk I/O

Docker

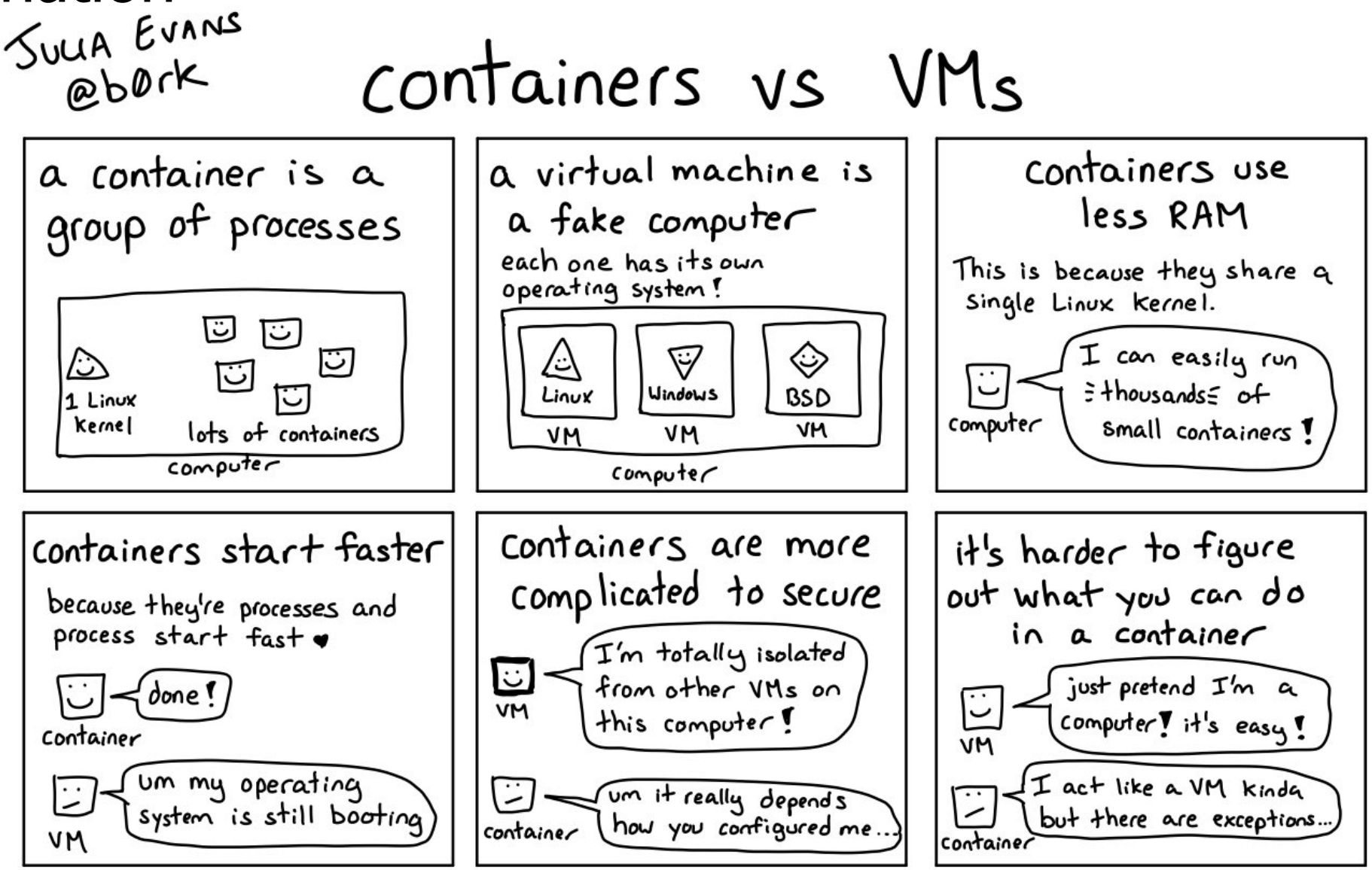
- Container packaging, distribution, and execution
  - Also created open standard for container runtimes
- Images
  - Describes starting state of a Docker container
  - Like a snapshot of the system
- Union file system
  - Image describes file system as a sequence of layers
    - Each layer includes some files
  - Overall file system is the *union* of all the layers
  - Layers can be reused in different images

Docker use cases

- Environments are hard to set up
  - Often the hardest part of starting software development
  - Containerized applications encapsulate requirements
  - Can be run on any system that has the same kernel it was built for

- Packages an application and its requirements into a container
  - Can be used by an individual to more easily run an application
  - Can be deployed to a cloud server to run

Bonus explanation



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